LECTURE NOTES

Nonlinear Solid Mechanics

Strong forms, weak forms, and consistent tangents

The path from finite-deformation kinematics to nonlinear finite elements rests on a single equilibrium statement. This note derives the reference and spatial forms, identifies work-conjugate strain and stress measures, and linearizes the residual to obtain Newton's tangent for finite element implementation.

Scope and conventions

A classical continuum is assumed. Unless otherwise stated, the discussion is quasi-static, the material is hyperelastic, and the prescribed reference loads are independent of deformation. Capital-coordinate derivatives and measures refer to the reference body $\mathcal{B}_0$; lowercase ones refer to the current deformed configuration $\mathcal{B}_t$.

1 Kinematics and Primary Solution Field

Let the reference body be $\mathcal{B}_0 \subset \mathbb{R}^3$ with boundary partition

$$\partial\mathcal{B}_0 = \Gamma_\varphi \cup \Gamma_T, \qquad \Gamma_\varphi \cap \Gamma_T = \varnothing.$$

The primary unknown is the deformation map, or equivalently the displacement:

$$\boldsymbol{\varphi} : \mathcal{B}_0 \to \mathbb{R}^3, \qquad \boldsymbol{u}(\boldsymbol{X}) = \boldsymbol{\varphi}(\boldsymbol{X}) - \boldsymbol{X}.$$

Its gradient and local volume ratio are

$$\boldsymbol{F} = \operatorname{Grad}\boldsymbol{\varphi} = \boldsymbol{I} + \operatorname{Grad}\boldsymbol{u}, \qquad J = \det\boldsymbol{F} > 0.$$

For a sufficiently smooth deformation, $J > 0$ ensures local orientation preservation and local invertibility.

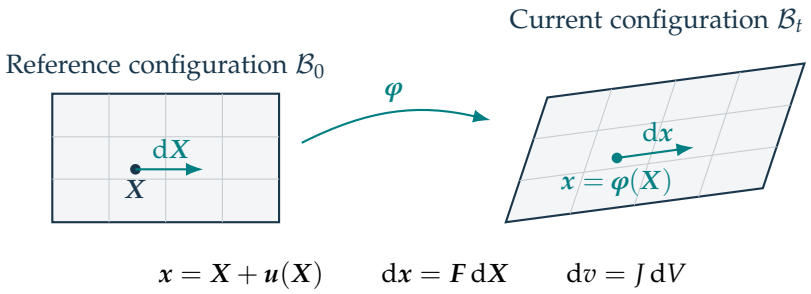


Figure 1. The deformation map moves material points, and its gradient maps local line elements. The illustrated affine deformation stretches and shears the body.

2 Stress Measures

In the reference configuration, the natural stress measure for the strong form is the first Piola–Kirchhoff stress $\boldsymbol{P}$. For a hyperelastic material with stored-energy density $\Psi(\boldsymbol{F}, \boldsymbol{X})$,

$$\boldsymbol{P} = \frac{\partial \Psi}{\partial \boldsymbol{F}}.$$

An *objective* hyperelastic law assigns the same stored energy under any superposed rigid motion: $\Psi(\boldsymbol{Q}\boldsymbol{F}, \boldsymbol{X}) = \Psi(\boldsymbol{F}, \boldsymbol{X})$ for every proper rotation $\boldsymbol{Q}$. Since $(\boldsymbol{Q}\boldsymbol{F})^T(\boldsymbol{Q}\boldsymbol{F}) = \boldsymbol{F}^T\boldsymbol{F}$, the energy per unit reference volume can be expressed in terms of the right Cauchy–Green tensor (Appendix A.1)

$$\boldsymbol{C} = \boldsymbol{F}^T\boldsymbol{F}.$$

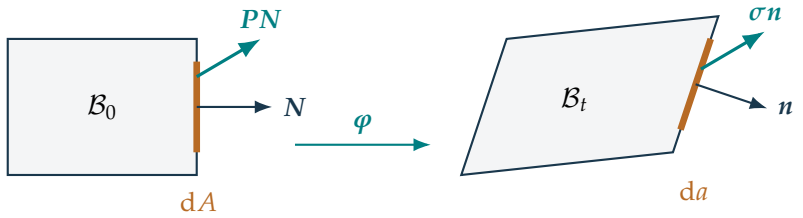
With this representation,

$$\boldsymbol{S} = 2\frac{\partial \Psi}{\partial \boldsymbol{C}}, \qquad \boldsymbol{P} = \boldsymbol{F}\boldsymbol{S},$$

where $\boldsymbol{S}$ is the second Piola–Kirchhoff stress.

The Cauchy stress $\boldsymbol{\sigma}$ is related to $\boldsymbol{P}$ by

$$\boldsymbol{P} = J\boldsymbol{\sigma}\boldsymbol{F}^{-T}.$$



$$\boldsymbol{n}\,\mathrm{d}a = J\boldsymbol{F}^{-T}\boldsymbol{N}\,\mathrm{d}A \qquad \boldsymbol{P}\boldsymbol{N}\,\mathrm{d}A = \boldsymbol{\sigma}\boldsymbol{n}\,\mathrm{d}a$$

Figure 2. Nanson's formula maps oriented surface area. Nominal and Cauchy tractions produce the same force on corresponding surface patches, but their force per unit area generally differs.

3 Strong Form in the Reference Configuration

For a *quasi-static* nonlinear elastic solid, loads and deformation may evolve, but inertia is neglected, so equilibrium is enforced at each loading stage. A *static* state is time-independent; both use the same equilibrium equation (Appendix A.2). The strong form is

$$\mathrm{Div}\,\boldsymbol{P} + \rho_0\boldsymbol{b} = \boldsymbol{0} \qquad \text{in } \mathcal{B}_0, \tag{1}$$

$$\boldsymbol{\varphi} = \overline{\boldsymbol{\varphi}} \qquad \text{on } \Gamma_\varphi, \tag{2}$$

$$\boldsymbol{P}\boldsymbol{N} = \overline{\boldsymbol{T}} \qquad \text{on } \Gamma_T. \tag{3}$$

Here, ρ_0 is the mass density per unit reference volume, $\boldsymbol{b}$ is the body force per unit mass, $\boldsymbol{N}$ is the outward unit normal in the reference configuration, and $\overline{\boldsymbol{T}}$ is the prescribed nominal traction per unit reference area.

Thus, the nonlinear boundary value problem is

$$\mathrm{Div}\,\boldsymbol{P}(\boldsymbol{F}, \boldsymbol{X}) + \rho_0\boldsymbol{b} = \boldsymbol{0} \qquad \text{in } \mathcal{B}_0, \tag{4}$$

$$\boldsymbol{\varphi} = \overline{\boldsymbol{\varphi}} \qquad \text{on } \Gamma_\varphi, \tag{5}$$

$$\boldsymbol{P}\boldsymbol{N} = \overline{\boldsymbol{T}} \qquad \text{on } \Gamma_T. \tag{6}$$

The deformation gradient is *affine* in the displacement gradient, i.e., linear plus the constant offset $\boldsymbol{I}$:

$$\boldsymbol{F} = \operatorname{Grad}\boldsymbol{\varphi} = \boldsymbol{I} + \operatorname{Grad}\boldsymbol{u}, \qquad \Delta\boldsymbol{F} = \operatorname{Grad}\Delta\boldsymbol{u}.$$

Thus, this kinematic relation is not nonlinear in $\boldsymbol{u}$; the nonlinearity arises from the quadratic strain terms and the stress response (Appendix A.3). The generally nonlinear stress response is encoded in the constitutive relation

$$\boldsymbol{P} = \boldsymbol{P}(\boldsymbol{F}, \boldsymbol{X}).$$

In a total Lagrangian description, Div acts on a fixed domain. Finite-deformation nonlinearity enters through $\boldsymbol{P}(\boldsymbol{F})$ (including nonlinear strain dependence and stress transformations) and, when present, through deformation-dependent loading.

4 Derivation of the Weak Form

Let $\boldsymbol{\eta}$ be an admissible virtual displacement field satisfying

$$\boldsymbol{\eta} = \boldsymbol{0} \qquad \text{on } \Gamma_\varphi.$$

The test space is

$$\mathcal{V}_0 = \left\{ \boldsymbol{\eta} \in [H^1(\mathcal{B}_0)]^3 : \boldsymbol{\eta} = \boldsymbol{0} \text{ on } \Gamma_\varphi \right\}.$$

The trial space is

$$\mathcal{V} = \left\{ \boldsymbol{\varphi} \in [H^1(\mathcal{B}_0)]^3 : \boldsymbol{\varphi} = \overline{\boldsymbol{\varphi}} \text{ on } \Gamma_\varphi \right\}.$$

These H^1 spaces describe the standard conforming finite-element setting. They guarantee square-integrable first derivatives but not finite energy for every nonlinear law: for example, quartic growth in $\boldsymbol{F}$ requires L^4 control of $\boldsymbol{F}$. Admissible solutions must also satisfy $J > 0$ and the integrability conditions required by the energy, stresses, and loads. The appropriate spaces and additional hypotheses in rigorous existence theory depend on the material law (Appendix A.4).

Start with the strong equilibrium equation

$$\operatorname{Div}\boldsymbol{P} + \rho_0\boldsymbol{b} = \boldsymbol{0}.$$

Multiply by an admissible virtual displacement $\boldsymbol{\eta}$ and integrate over the reference body:

$$\int_{\mathcal{B}_0} \boldsymbol{\eta} \cdot (\operatorname{Div}\boldsymbol{P} + \rho_0\boldsymbol{b}) \, \mathrm{d}V = 0.$$

Thus

$$\int_{\mathcal{B}_0} \boldsymbol{\eta} \cdot \operatorname{Div}\boldsymbol{P} \, \mathrm{d}V + \int_{\mathcal{B}_0} \rho_0\boldsymbol{b} \cdot \boldsymbol{\eta} \, \mathrm{d}V = 0.$$

Using the divergence theorem,

$$\int_{\mathcal{B}_0} \boldsymbol{\eta} \cdot \operatorname{Div}\boldsymbol{P} \, \mathrm{d}V = \int_{\partial\mathcal{B}_0} \boldsymbol{\eta} \cdot \boldsymbol{P}\boldsymbol{N} \, \mathrm{d}A - \int_{\mathcal{B}_0} \boldsymbol{P} : \operatorname{Grad}\boldsymbol{\eta} \, \mathrm{d}V.$$

Substituting and rearranging gives

$$\int_{\mathcal{B}_0} \boldsymbol{P} : \operatorname{Grad}\boldsymbol{\eta} \, \mathrm{d}V = \int_{\mathcal{B}_0} \rho_0\boldsymbol{b} \cdot \boldsymbol{\eta} \, \mathrm{d}V + \int_{\partial\mathcal{B}_0} \boldsymbol{\eta} \cdot \boldsymbol{P}\boldsymbol{N} \, \mathrm{d}A.$$

Now split the boundary:

$$\partial\mathcal{B}_0 = \Gamma_\varphi \cup \Gamma_T.$$

Since $\boldsymbol{\eta} = \mathbf{0}$ on Γ_φ, the essential-boundary contribution vanishes. On Γ_T,

$$\boldsymbol{P}\boldsymbol{N} = \overline{\boldsymbol{T}}.$$

Therefore, the weak form is

$$\int_{\mathcal{B}_0} \boldsymbol{P}(\boldsymbol{F}) : \operatorname{Grad}\boldsymbol{\eta}\,\mathrm{d}V = \int_{\mathcal{B}_0} \rho_0\boldsymbol{b}\cdot\boldsymbol{\eta}\,\mathrm{d}V + \int_{\Gamma_T} \overline{\boldsymbol{T}}\cdot\boldsymbol{\eta}\,\mathrm{d}A \tag{7}$$

for all admissible virtual displacements $\boldsymbol{\eta} \in \mathcal{V}_0$.

Equivalently, define the residual functional

$$\mathcal{R}(\boldsymbol{\varphi};\boldsymbol{\eta}) = \int_{\mathcal{B}_0} \boldsymbol{P}(\operatorname{Grad}\boldsymbol{\varphi}) : \operatorname{Grad}\boldsymbol{\eta}\,\mathrm{d}V - \int_{\mathcal{B}_0} \rho_0\boldsymbol{b}\cdot\boldsymbol{\eta}\,\mathrm{d}V - \int_{\Gamma_T} \overline{\boldsymbol{T}}\cdot\boldsymbol{\eta}\,\mathrm{d}A.$$

Then the weak form is

$$\text{Find } \boldsymbol{\varphi} \in \mathcal{V} \text{ such that } \mathcal{R}(\boldsymbol{\varphi};\boldsymbol{\eta}) = 0 \quad \forall\boldsymbol{\eta} \in \mathcal{V}_0.$$

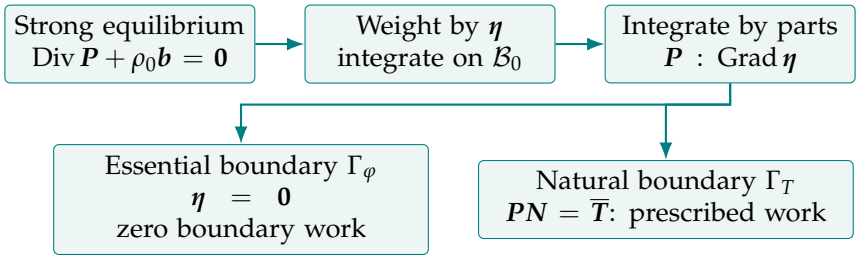


Figure 3. Integration by parts transfers one derivative to the test field, revealing boundary work. Essential data constrain admissible fields; natural data enter the load functional.

5 Interpretation as Virtual Work

The weak form can be written as

$$\delta W_{\text{int}} = \delta W_{\text{ext}}.$$

Here

$$\delta W_{\text{int}} = \int_{\mathcal{B}_0} \boldsymbol{P} : \delta\boldsymbol{F}\,\mathrm{d}V, \qquad \delta\boldsymbol{F} = \operatorname{Grad}\boldsymbol{\eta},$$

and

$$\delta W_{\text{ext}} = \int_{\mathcal{B}_0} \rho_0\boldsymbol{b}\cdot\boldsymbol{\eta}\,\mathrm{d}V + \int_{\Gamma_T} \overline{\boldsymbol{T}}\cdot\boldsymbol{\eta}\,\mathrm{d}A.$$

Thus

$$\int_{\mathcal{B}_0} \boldsymbol{P} : \delta\boldsymbol{F}\,\mathrm{d}V = \int_{\mathcal{B}_0} \rho_0\boldsymbol{b}\cdot\boldsymbol{\eta}\,\mathrm{d}V + \int_{\Gamma_T} \overline{\boldsymbol{T}}\cdot\boldsymbol{\eta}\,\mathrm{d}A.$$

The key point is that $\boldsymbol{\eta}$ is arbitrary except for satisfying the homogeneous essential boundary condition. Therefore, the weak statement enforces equilibrium in an integral sense over the body, and the natural traction boundary condition arises through integration by parts.

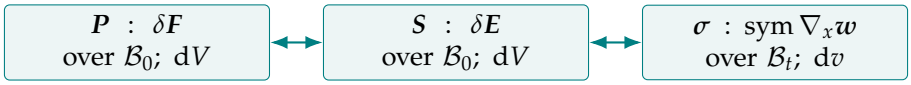


Figure 4. Three equivalent representations of internal virtual work. The spatial work density must be paired with the current volume measure.

6 Hyperelastic Variational Form

For a hyperelastic material under conservative loading, equilibrium is a stationarity condition. The explicit expression below assumes *dead loads*: $\boldsymbol{b}$ and $\overline{\boldsymbol{T}}$ are prescribed independently of the deformation in reference coordinates. Define

$$\Pi(\boldsymbol{\varphi}) = \int_{\mathcal{B}_0} \Psi(\operatorname{Grad}\boldsymbol{\varphi}, \boldsymbol{X})\,\mathrm{d}V - \int_{\mathcal{B}_0} \rho_0 \boldsymbol{b}\cdot\boldsymbol{\varphi}\,\mathrm{d}V - \int_{\Gamma_T} \overline{\boldsymbol{T}}\cdot\boldsymbol{\varphi}\,\mathrm{d}A.$$

The equilibrium deformation satisfies

$$\delta\Pi(\boldsymbol{\varphi};\boldsymbol{\eta}) = 0 \qquad \forall \boldsymbol{\eta} \in \mathcal{V}_0.$$

Replacing $\boldsymbol{\varphi}$ by $\boldsymbol{u}$ in the dead-load potential changes only a constant. For other conservative loads, the appropriate load potential must be used. Stationarity alone does not imply a minimum or stability.

Taking the first variation,

$$\delta\Pi = \int_{\mathcal{B}_0} \frac{\partial\Psi}{\partial\boldsymbol{F}} : \delta\boldsymbol{F}\,\mathrm{d}V - \int_{\mathcal{B}_0} \rho_0\boldsymbol{b}\cdot\boldsymbol{\eta}\,\mathrm{d}V - \int_{\Gamma_T} \overline{\boldsymbol{T}}\cdot\boldsymbol{\eta}\,\mathrm{d}A.$$

Since

$$\boldsymbol{P} = \frac{\partial\Psi}{\partial\boldsymbol{F}}, \qquad \delta\boldsymbol{F} = \operatorname{Grad}\boldsymbol{\eta},$$

we recover

$$\int_{\mathcal{B}_0} \boldsymbol{P} : \operatorname{Grad}\boldsymbol{\eta}\,\mathrm{d}V = \int_{\mathcal{B}_0} \rho_0\boldsymbol{b}\cdot\boldsymbol{\eta}\,\mathrm{d}V + \int_{\Gamma_T} \overline{\boldsymbol{T}}\cdot\boldsymbol{\eta}\,\mathrm{d}A.$$

Thus, for conservative hyperelasticity,

$$\text{weak equilibrium} \quad\Longleftrightarrow\quad \text{stationarity of total potential energy.}$$

7 Spatial Form

The equilibrium equation can be expressed in the current configuration $\mathcal{B}_t$. The strong form is

$$\operatorname{div}\boldsymbol{\sigma} + \rho\boldsymbol{b} = \boldsymbol{0} \qquad \text{in } \mathcal{B}_t,$$

with the traction condition

$$\boldsymbol{\sigma}\boldsymbol{n} = \bar{\boldsymbol{t}} \qquad \text{on } \gamma_t.$$

The corresponding weak form is

$$\int_{\mathcal{B}_t} \boldsymbol{\sigma} : \nabla_x \boldsymbol{w}\,\mathrm{d}v = \int_{\mathcal{B}_t} \rho\boldsymbol{b}\cdot\boldsymbol{w}\,\mathrm{d}v + \int_{\gamma_t} \bar{\boldsymbol{t}}\cdot\boldsymbol{w}\,\mathrm{d}a$$

Here, $\boldsymbol{w} = \boldsymbol{\eta} \circ \boldsymbol{\varphi}^{-1}$ is the spatial virtual displacement, vanishing on the deformed essential boundary. Composition means

$$\boldsymbol{w}(\boldsymbol{x}) = \boldsymbol{\eta}\big(\boldsymbol{\varphi}^{-1}(\boldsymbol{x})\big), \qquad \boldsymbol{w}(\boldsymbol{\varphi}(\boldsymbol{X})) = \boldsymbol{\eta}(\boldsymbol{X}).$$

It assigns the same virtual displacement to corresponding current and reference points; the inverse assumes a one-to-one deformation (Appendix A.5). If $\boldsymbol{w}$ is instead interpreted as a virtual velocity, the same expression is a virtual-power balance. The configurations are related by

$$\operatorname{Grad} \boldsymbol{\eta} = (\nabla_x \boldsymbol{w})\boldsymbol{F}, \qquad \mathrm{d}v = J\,\mathrm{d}V, \qquad \rho J = \rho_0.$$

Together with $\boldsymbol{P} = J\boldsymbol{\sigma}\boldsymbol{F}^{-T}$, these transformations yield identical internal virtual work.

When angular momentum balance is enforced, the Cauchy stress is symmetric, $\boldsymbol{\sigma} = \boldsymbol{\sigma}^T$, and

$$\boldsymbol{\sigma} : \nabla_x \boldsymbol{w} = \boldsymbol{\sigma} : \operatorname{sym} \nabla_x \boldsymbol{w}.$$

Therefore, the spatial weak form is often written as

$$\int_{\mathcal{B}_t} \boldsymbol{\sigma} : \boldsymbol{\varepsilon}(\boldsymbol{w})\,\mathrm{d}v = \int_{\mathcal{B}_t} \rho \boldsymbol{b} \cdot \boldsymbol{w}\,\mathrm{d}v + \int_{\gamma_t} \bar{\boldsymbol{t}} \cdot \boldsymbol{w}\,\mathrm{d}a,$$

where

$$\boldsymbol{\varepsilon}(\boldsymbol{w}) = \operatorname{sym} \nabla_x \boldsymbol{w}.$$

This resembles the small-strain weak form, but it is not the same problem because the domain $\mathcal{B}_t$, stress $\boldsymbol{\sigma}$, and deformation are configuration-dependent.

8 Dynamic Extension

For finite-deformation dynamics, the reference strong form is

$$\operatorname{Div} \boldsymbol{P} + \rho_0 \boldsymbol{b} = \rho_0 \ddot{\boldsymbol{\varphi}} \qquad \text{in } \mathcal{B}_0.$$

The weak form is

$$\int_{\mathcal{B}_0} \rho_0 \ddot{\boldsymbol{\varphi}} \cdot \boldsymbol{\eta}\,\mathrm{d}V + \int_{\mathcal{B}_0} \boldsymbol{P} : \operatorname{Grad} \boldsymbol{\eta}\,\mathrm{d}V = \int_{\mathcal{B}_0} \rho_0 \boldsymbol{b} \cdot \boldsymbol{\eta}\,\mathrm{d}V + \int_{\Gamma_T} \overline{\boldsymbol{T}} \cdot \boldsymbol{\eta}\,\mathrm{d}A.$$

This serves as a natural starting point for nonlinear transient finite element formulations.

9 Finite Elements

Use the same shape functions to interpolate the reference geometry and the displacement:

$$\boldsymbol{X} = \sum_A N_A \boldsymbol{X}_A, \qquad \boldsymbol{u}^h = \sum_A N_A \boldsymbol{d}_A, \qquad \boldsymbol{\varphi}_A = \boldsymbol{X}_A + \boldsymbol{d}_A.$$

This geometry-reproduction condition makes motion and displacement interpolation equivalent. The deformation is approximated as

$$\boldsymbol{\varphi}^h(\boldsymbol{X}) = \sum_{A=1}^{n_{\text{node}}} N_A(\boldsymbol{X}) \boldsymbol{\varphi}_A.$$

Then

$$\boldsymbol{F}^h = \operatorname{Grad} \boldsymbol{\varphi}^h = \sum_A \boldsymbol{\varphi}_A \otimes \operatorname{Grad} N_A.$$

The virtual displacement is approximated as

$$\boldsymbol{\eta}^h = \sum_A N_A \boldsymbol{\eta}_A.$$

Substituting into the weak form yields the nonlinear algebraic residual

$$\boldsymbol{R}_A = \int_{\mathcal{B}_0} \boldsymbol{P}(\boldsymbol{F}^h)\,\mathrm{Grad}\,N_A\,\mathrm{d}V - \int_{\mathcal{B}_0} \rho_0 \boldsymbol{b} N_A\,\mathrm{d}V - \int_{\Gamma_T} \overline{\boldsymbol{T}} N_A\,\mathrm{d}A = \boldsymbol{0}.$$

After assembly and imposition of the essential boundary conditions, the equations for the free degrees of freedom are

$$\boldsymbol{R}(\boldsymbol{d}) = \boldsymbol{0},$$

where $\boldsymbol{d}$ collects the nodal displacement unknowns.

Because

$$\boldsymbol{P} = \boldsymbol{P}(\boldsymbol{F}^h), \qquad \boldsymbol{F}^h = \boldsymbol{I} + \mathrm{Grad}\,\boldsymbol{u}^h,$$

the residual is nonlinear in the nodal unknowns. Newton's method iterates as

$$\boldsymbol{K}_T(\boldsymbol{d}^{(k)})\Delta\boldsymbol{d}^{(k)} = -\boldsymbol{R}(\boldsymbol{d}^{(k)}),$$

where

$$\boldsymbol{K}_T = \frac{\partial \boldsymbol{R}}{\partial \boldsymbol{d}}$$

is the tangent stiffness matrix.

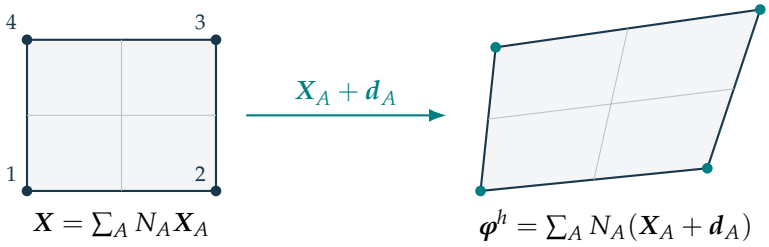


Figure 5. Isoparametric interpolation on a four-node quadrilateral (two-dimensional illustration). Shared shape functions reproduce the reference geometry and ensure that $\boldsymbol{F}^h = \boldsymbol{I} + \mathrm{Grad}\,\boldsymbol{u}^h$ is consistent with motion interpolation.

10 Work-Conjugate Stress and Strain

The Green–Lagrange strain is

$$\boldsymbol{E} = \frac{1}{2}(\boldsymbol{C} - \boldsymbol{I}) = \frac{1}{2}(\boldsymbol{F}^T\boldsymbol{F} - \boldsymbol{I}).$$

For a hyperelastic material with $\Psi = \Psi(\boldsymbol{E})$ or $\Psi = \Psi(\boldsymbol{C})$, the second Piola–Kirchhoff stress is

$$\boldsymbol{S} = \frac{\partial \Psi}{\partial \boldsymbol{E}} = 2\frac{\partial \Psi}{\partial \boldsymbol{C}}.$$

The factor 2 follows from $\delta\boldsymbol{E} = \frac{1}{2}\delta\boldsymbol{C}$: the energy variation is $\delta\Psi = \boldsymbol{S} : \delta\boldsymbol{E} = \frac{1}{2}\boldsymbol{S} : \delta\boldsymbol{C}$. Here $\Psi(\boldsymbol{E})$ and $\Psi(\boldsymbol{C})$ denote equivalent representations of the same energy, with their arguments changed consistently (Appendix A.6).

The first Piola–Kirchhoff stress is

$$\boldsymbol{P} = \boldsymbol{F}\boldsymbol{S}.$$

The virtual Green–Lagrange strain is

$$\delta\boldsymbol{E} = \frac{1}{2}\left(\delta\boldsymbol{F}^T\boldsymbol{F} + \boldsymbol{F}^T\delta\boldsymbol{F}\right), \qquad \delta\boldsymbol{F} = \operatorname{Grad}\boldsymbol{\eta}.$$

Since $\boldsymbol{S}$ is symmetric,

$$\boldsymbol{S} : \delta\boldsymbol{E} = \boldsymbol{S} : \left(\boldsymbol{F}^T\delta\boldsymbol{F}\right) = \boldsymbol{F}\boldsymbol{S} : \delta\boldsymbol{F} = \boldsymbol{P} : \delta\boldsymbol{F}.$$

Therefore, the internal virtual work may be expressed equivalently as

$$\delta W_{\text{int}} = \int_{\mathcal{B}_0} \boldsymbol{P} : \delta\boldsymbol{F}\,\mathrm{d}V = \int_{\mathcal{B}_0} \boldsymbol{S} : \delta\boldsymbol{E}\,\mathrm{d}V.$$

The weak form is

$$\int_{\mathcal{B}_0} \boldsymbol{S} : \delta\boldsymbol{E}\,\mathrm{d}V = \int_{\mathcal{B}_0} \rho_0\boldsymbol{b}\cdot\boldsymbol{\eta}\,\mathrm{d}V + \int_{\Gamma_T} \overline{\boldsymbol{T}}\cdot\boldsymbol{\eta}\,\mathrm{d}A \qquad \forall\boldsymbol{\eta}\in\mathcal{V}_0.$$

This is the classical total Lagrangian form. It is especially useful because all integrations are performed over the reference configuration, and the stress $\boldsymbol{S}$ is work-conjugate to the strain $\boldsymbol{E}$.

For the displacement field,

$$\boldsymbol{F} = \boldsymbol{I} + \operatorname{Grad}\boldsymbol{u},$$

and

$$\boldsymbol{E} = \frac{1}{2}\left(\operatorname{Grad}\boldsymbol{u} + \operatorname{Grad}\boldsymbol{u}^T + \operatorname{Grad}\boldsymbol{u}^T\operatorname{Grad}\boldsymbol{u}\right).$$

The virtual strain is

$$\delta\boldsymbol{E} = \operatorname{sym}\operatorname{Grad}\boldsymbol{\eta} + \frac{1}{2}\left(\operatorname{Grad}\boldsymbol{\eta}^T\operatorname{Grad}\boldsymbol{u} + \operatorname{Grad}\boldsymbol{u}^T\operatorname{Grad}\boldsymbol{\eta}\right).$$

Equivalently,

$$\delta\boldsymbol{E} = \operatorname{sym}\left(\boldsymbol{F}^T\operatorname{Grad}\boldsymbol{\eta}\right).$$

Thus, the difference between the $\boldsymbol{P}$–$\boldsymbol{F}$ and $\boldsymbol{S}$–$\boldsymbol{E}$ forms is not a different physical statement; it is a different work-conjugate representation of the same weak equilibrium equation.

11 Consistent Linearization and Tangent Stiffness

Consider the residual functional

$$\mathcal{R}(\boldsymbol{\varphi};\boldsymbol{\eta}) = \int_{\mathcal{B}_0} \boldsymbol{P}(\boldsymbol{F}) : \operatorname{Grad}\boldsymbol{\eta}\,\mathrm{d}V - \int_{\mathcal{B}_0} \rho_0\boldsymbol{b}\cdot\boldsymbol{\eta}\,\mathrm{d}V - \int_{\Gamma_T} \overline{\boldsymbol{T}}\cdot\boldsymbol{\eta}\,\mathrm{d}A.$$

Newton's method requires the directional derivative of the residual with respect to an incremental displacement $\Delta\boldsymbol{u}$ or deformation $\Delta\boldsymbol{\varphi}$.

Let

$$\Delta\boldsymbol{F} = \operatorname{Grad}\Delta\boldsymbol{u}.$$

The constitutive tangent associated with the first Piola–Kirchhoff stress is the fourth-order tensor

$$\mathbb{A} = \frac{\partial\boldsymbol{P}}{\partial\boldsymbol{F}}.$$

Therefore,

$$\Delta \boldsymbol{P} = \mathbb{A} : \Delta \boldsymbol{F}.$$

If the external loads are deformation-independent, the consistent tangent is

$$D\mathcal{R}(\boldsymbol{\varphi}; \boldsymbol{\eta})[\Delta \boldsymbol{u}] = \int_{\mathcal{B}_0} \operatorname{Grad} \boldsymbol{\eta} : \mathbb{A} : \operatorname{Grad} \Delta \boldsymbol{u} \, \mathrm{d}V.$$

The Newton correction solves

$$D\mathcal{R}(\boldsymbol{\varphi}^{(k)}; \boldsymbol{\eta})[\Delta \boldsymbol{u}^{(k)}] = -\mathcal{R}(\boldsymbol{\varphi}^{(k)}; \boldsymbol{\eta}) \qquad \forall \boldsymbol{\eta} \in \mathcal{V}_0.$$

In a finite element discretization, this becomes

$$\boldsymbol{K}_T(\boldsymbol{d}^{(k)}) \Delta \boldsymbol{d}^{(k)} = -\boldsymbol{R}(\boldsymbol{d}^{(k)}).$$

The term *consistent* means that the tangent is obtained by exact linearization of the residual used in the Newton iteration. This yields quadratic convergence near the solution when the other hypotheses for Newton convergence are satisfied.

For the total Lagrangian ***S*–*E*** formulation, the internal virtual work is

$$\mathcal{G}_{\text{int}}(\boldsymbol{u}; \boldsymbol{\eta}) = \int_{\mathcal{B}_0} \boldsymbol{S} : \delta \boldsymbol{E} \, \mathrm{d}V.$$

Its linearization has two contributions:

$$D\mathcal{G}_{\text{int}}[\Delta \boldsymbol{u}] = \int_{\mathcal{B}_0} \delta \boldsymbol{E} : \mathbb{C} : \Delta \boldsymbol{E} \, \mathrm{d}V + \int_{\mathcal{B}_0} \boldsymbol{S} : \Delta(\delta \boldsymbol{E}) \, \mathrm{d}V,$$

where

$$\mathbb{C} = \frac{\partial \boldsymbol{S}}{\partial \boldsymbol{E}}$$

is the material elasticity tensor,

$$\Delta \boldsymbol{E} = \operatorname{sym}(\boldsymbol{F}^T \operatorname{Grad} \Delta \boldsymbol{u}),$$

and

$$\Delta(\delta \boldsymbol{E}) = \operatorname{sym}\left((\operatorname{Grad} \Delta \boldsymbol{u})^T \operatorname{Grad} \boldsymbol{\eta} \right).$$

Thus

$$D\mathcal{G}_{\text{int}}[\Delta \boldsymbol{u}] = \int_{\mathcal{B}_0} \delta \boldsymbol{E} : \mathbb{C} : \Delta \boldsymbol{E} \, \mathrm{d}V + \int_{\mathcal{B}_0} \boldsymbol{S} : \operatorname{sym}\left((\operatorname{Grad} \Delta \boldsymbol{u})^T \operatorname{Grad} \boldsymbol{\eta} \right) \, \mathrm{d}V.$$

The first term gives the material stiffness; the second gives the geometric or initial-stress stiffness. Both are already contained in $\mathbb{A} = \partial \boldsymbol{P} / \partial \boldsymbol{F}$: a separate geometric term must not be added again to the ***P*–*F*** linearization.

With spatial indices i, j and material indices I, J, K, L, the connection is

$$A_{iIjJ} = \delta_{ij} S_{JI} + F_{iK} C_{KIJL} F_{jL}, \qquad (K_{\text{int}})_{Ai,Bj} = \int_{\mathcal{B}_0} N_{A,I} A_{iIjJ} N_{B,J} \, \mathrm{d}V.$$

Repeated indices are summed. For a twice continuously differentiable hyperelastic energy, the internal tangent has major symmetry; it need not be positive definite.

For follower loads, contact tractions, pressure loads acting normal to the deformed surface, or other deformation-dependent loads, the external virtual work must also be linearized. Then

$$\boldsymbol{K}_T = \boldsymbol{K}_{\text{int}} - \boldsymbol{K}_{\text{ext,load}}.$$

This load stiffness is essential for correct convergence and stability in large-deformation problems.

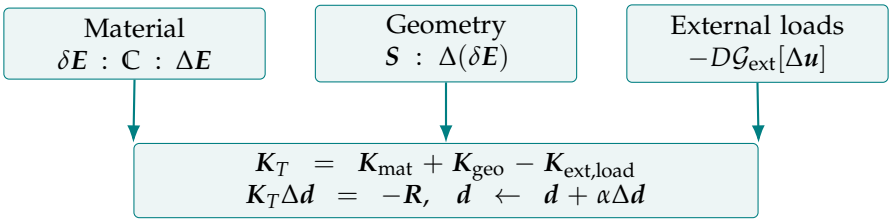


Figure 6. The consistent tangent differentiates the residual. Dead loads have a zero load tangent. A line search may choose $0 < \alpha \leq 1$; quadratic local convergence requires the full Newton step and the usual regularity and nonsingularity assumptions.

12 Reduction to Small-Strain Linear Elasticity

The finite-deformation weak form reduces to the small-strain weak form when the displacement gradients are small:

$$\|\operatorname{Grad} \boldsymbol{u}\| \ll 1.$$

Then

$$\boldsymbol{F} = \boldsymbol{I} + \operatorname{Grad} \boldsymbol{u} \approx \boldsymbol{I},$$

and the Green–Lagrange strain becomes

$$\boldsymbol{E} = \frac{1}{2}\left(\operatorname{Grad} \boldsymbol{u} + \operatorname{Grad} \boldsymbol{u}^T + \operatorname{Grad} \boldsymbol{u}^T \operatorname{Grad} \boldsymbol{u}\right) \approx \frac{1}{2}\left(\operatorname{Grad} \boldsymbol{u} + \operatorname{Grad} \boldsymbol{u}^T\right) = \boldsymbol{\varepsilon}(\boldsymbol{u}).$$

Similarly,

$$\delta \boldsymbol{E} = \operatorname{sym}(\boldsymbol{F}^T \operatorname{Grad} \boldsymbol{\eta}) \approx \operatorname{sym} \operatorname{Grad} \boldsymbol{\eta} = \boldsymbol{\varepsilon}(\boldsymbol{\eta}).$$

When linearizing about an unstressed reference state with small displacement gradients, the stress measures agree to first order:

$$\boldsymbol{P} \approx \boldsymbol{S} \approx \boldsymbol{\sigma}.$$

For a linear elastic material,

$$\boldsymbol{\sigma} = \mathbb{C} : \boldsymbol{\varepsilon}(\boldsymbol{u}),$$

where $\mathbb{C}$ is the fourth-order linear elastic stiffness tensor.

Therefore, the finite-deformation weak form

$$\int_{\mathcal{B}_0} \boldsymbol{S} : \delta \boldsymbol{E} \, \mathrm{d}V = \int_{\mathcal{B}_0} \rho_0 \boldsymbol{b} \cdot \boldsymbol{\eta} \, \mathrm{d}V + \int_{\Gamma_T} \overline{\boldsymbol{T}} \cdot \boldsymbol{\eta} \, \mathrm{d}A$$

reduces to

$$\int_{\Omega} \boldsymbol{\varepsilon}(\boldsymbol{\eta}) : \mathbb{C} : \boldsymbol{\varepsilon}(\boldsymbol{u}) \, \mathrm{d}V = \int_{\Omega} \boldsymbol{f} \cdot \boldsymbol{\eta} \, \mathrm{d}V + \int_{\Gamma_t} \bar{\boldsymbol{t}} \cdot \boldsymbol{\eta} \, \mathrm{d}A \qquad \forall \boldsymbol{\eta} \in \mathcal{V}_0.$$

Here, $\boldsymbol{f} = \rho_0 \boldsymbol{b}$ to the order retained, and Ω denotes the common reference/current domain in this limit. This is the familiar weak form of small-strain linear elasticity. Small strain alone is insufficient if rotations are large; linearization about a prestressed state can retain geometric stiffness.

The corresponding strong form is

$$\begin{aligned} \operatorname{Div} \boldsymbol{\sigma} + \boldsymbol{f} &= \boldsymbol{0} \quad \text{in } \Omega, \\ \boldsymbol{u} &= \bar{\boldsymbol{u}} \quad \text{on } \Gamma_u, \\ \boldsymbol{\sigma} \boldsymbol{n} &= \bar{\boldsymbol{t}} \quad \text{on } \Gamma_t. \end{aligned}$$

Here,

$$\boldsymbol{\varepsilon}(\boldsymbol{u}) = \operatorname{sym} \nabla \boldsymbol{u}, \qquad \boldsymbol{\sigma} = \mathbb{C} : \boldsymbol{\varepsilon}(\boldsymbol{u}).$$

The key limiting assumptions are:

(i) the reference and current configurations are nearly identical;

(ii) Grad $\boldsymbol{u}$ is sufficiently small that quadratic terms can be neglected;

(iii) stress measures coincide to first order;

(iv) the constitutive model is linearized about the reference configuration;

(v) load directions and boundary normals are treated as fixed rather than deformation-dependent.

Under these assumptions, finite-deformation virtual work reduces to the classical small-strain virtual work statement.



Rigid rotation: $\boldsymbol{F} = \boldsymbol{Q}$
$\boldsymbol{E} = \boldsymbol{0}$, but Grad $\boldsymbol{u}$ need not be small

Small displacement gradients
$\boldsymbol{E} \approx \operatorname{sym} \operatorname{Grad} \boldsymbol{u}$

Figure 7. Small strain and small displacement gradient are distinct assumptions. A finite rigid rotation has zero Green–Lagrange strain but is not exactly represented by infinitesimal strain.

13 Summary

The boundary value problem for nonlinear solid mechanics in the reference configuration is

$$\operatorname{Div} \boldsymbol{P} + \rho_0 \boldsymbol{b} = \boldsymbol{0}, \qquad \boldsymbol{\varphi} = \overline{\boldsymbol{\varphi}} \text{ on } \Gamma_\varphi, \qquad \boldsymbol{P}\boldsymbol{N} = \overline{\boldsymbol{T}} \text{ on } \Gamma_T.$$

Its weak form is

$$\int_{\mathcal{B}_0} \boldsymbol{P} : \operatorname{Grad} \boldsymbol{\eta} \, \mathrm{d}V = \int_{\mathcal{B}_0} \rho_0 \boldsymbol{b} \cdot \boldsymbol{\eta} \, \mathrm{d}V + \int_{\Gamma_T} \overline{\boldsymbol{T}} \cdot \boldsymbol{\eta} \, \mathrm{d}A \qquad \forall \boldsymbol{\eta} \in \mathcal{V}_0.$$

Equivalently,

$$\int_{\mathcal{B}_0} \boldsymbol{S} : \delta \boldsymbol{E} \, \mathrm{d}V = \int_{\mathcal{B}_0} \rho_0 \boldsymbol{b} \cdot \boldsymbol{\eta} \, \mathrm{d}V + \int_{\Gamma_T} \overline{\boldsymbol{T}} \cdot \boldsymbol{\eta} \, \mathrm{d}A.$$

The first form uses the work-conjugate pair $(\boldsymbol{P}, \boldsymbol{F})$, while the second uses the work-conjugate pair $(\boldsymbol{S}, \boldsymbol{E})$. Linearizing the nonlinear residual yields the consistent tangent stiffness used in Newton iteration. In the small-displacement-gradient limit, the formulation reduces to the standard weak form of small-strain linear elasticity.

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A Questions and Answers: Constitutive and Mathematical Qualifications

This appendix provides further detail on the six qualifications highlighted in the main text. Throughout, Ψ denotes the stored energy per unit reference volume, and tensor derivatives use the double-contraction convention $\boldsymbol{A} : \boldsymbol{B} = \sum_{i,j} A_{ij} B_{ij}$.

A.1 What is an objective hyperelastic law?

Answer. Objectivity, or material frame indifference, means that superimposing a rigid translation or rotation on the deformed body does not change its stored energy. Such a motion changes the body's placement but not its elastic distortion.

Let the deformation be $\boldsymbol{x} = \boldsymbol{\varphi}(\boldsymbol{X})$, and consider the superposition

$$\boldsymbol{\varphi}^*(\boldsymbol{X}) = \boldsymbol{c} + \boldsymbol{Q}\boldsymbol{\varphi}(\boldsymbol{X}), \qquad \boldsymbol{Q}^T\boldsymbol{Q} = \boldsymbol{I}, \quad \det \boldsymbol{Q} = 1.$$

The translation $\boldsymbol{c}$ and rotation $\boldsymbol{Q}$ are spatially uniform and may depend on time. Differentiation with respect to $\boldsymbol{X}$ yields $\boldsymbol{F}^* = \boldsymbol{Q}\boldsymbol{F}$. Objectivity requires

$$\boxed{\Psi(\boldsymbol{Q}\boldsymbol{F}, \boldsymbol{X}) = \Psi(\boldsymbol{F}, \boldsymbol{X}).}$$

Since

$$\boldsymbol{C}^* = (\boldsymbol{F}^*)^T\boldsymbol{F}^* = \boldsymbol{F}^T\boldsymbol{Q}^T\boldsymbol{Q}\boldsymbol{F} = \boldsymbol{C},$$

an objective energy can be expressed as a function of $\boldsymbol{C}$ and the material position $\boldsymbol{X}$. The superposed rotation drops out, while information about stretch and distortion remains.

For example, rotating an already stretched rubber specimen changes neither its stretch nor its stored energy. Its stress state rotates with it: $\boldsymbol{\sigma}^* = \boldsymbol{Q}\boldsymbol{\sigma}\boldsymbol{Q}^T$.

Objectivity does not imply isotropy. An objective material may have preferred fiber directions. Objectivity concerns a rigid rotation of the entire deformed body, whereas isotropy concerns the absence of preferred material directions. Reference fiber directions can remain parameters of the energy expressed in terms of $\boldsymbol{C}$.

A.2 How does quasi-static differ from static?

Answer. A static state is a time-independent equilibrium. A quasi-static process is a sequence of equilibrium states that evolves, with inertial effects neglected. In either case, the retained equilibrium equation is

$$\operatorname{Div} \boldsymbol{P} + \rho_0 \boldsymbol{b} = \boldsymbol{0}.$$

In a quasi-static process, $\boldsymbol{P}$, $\boldsymbol{b}$, the prescribed boundary data, and $\boldsymbol{\varphi}$ may vary with time or a loading parameter. The body can move: its acceleration is not necessarily exactly zero. The approximation is to neglect the inertial term in the full balance

$$\operatorname{Div} \boldsymbol{P} + \rho_0 \boldsymbol{b} = \rho_0 \ddot{\boldsymbol{\varphi}}.$$

A stretched specimen held at an equilibrium extension is static. Stretching it slowly through increasing extensions is approximately quasi-static. Suddenly pulling or releasing it generally requires a dynamic analysis.

For rate-independent hyperelasticity, quasi-static "time" may simply parameterize the loading. Thus, *static* suffices for a single equilibrium state, whereas *quasi-static* emphasizes the loading process. Slow external loading alone does not ensure negligible inertia: an instability such as snap-through can trigger rapid motion.

A.3 Why emphasize that the deformation gradient is affine?

Answer. An affine map is linear plus a constant offset. With $\boldsymbol{H} = \operatorname{Grad} \boldsymbol{u}$,

$$\boldsymbol{F}(\boldsymbol{H}) = \boldsymbol{I} + \boldsymbol{H}.$$

The offset is $\boldsymbol{I}$, so this map is not strictly linear: $\boldsymbol{F}(\boldsymbol{0}) = \boldsymbol{I} \neq \boldsymbol{0}$. However, its increment is linear,

$$\Delta \boldsymbol{F} = \operatorname{Grad} \Delta \boldsymbol{u},$$

and its second derivative with respect to displacement vanishes. Thus, forming $\boldsymbol{F}$ from $\boldsymbol{u}$ does not introduce a nonlinear dependence.

By comparison, the Green–Lagrange strain contains a quadratic term:

$$\boldsymbol{E} = \frac{1}{2}(\boldsymbol{H} + \boldsymbol{H}^T + \boldsymbol{H}^T \boldsymbol{H}).$$

Even a constitutive law linear in $\boldsymbol{E}$, such as $\boldsymbol{S} = \mathbb{C} : \boldsymbol{E}$, generally yields a nonlinear displacement problem because

$$\boldsymbol{P} = (\boldsymbol{I} + \boldsymbol{H})\big[\mathbb{C} : \boldsymbol{E}(\boldsymbol{H})\big].$$

The term *affine* indicates where the nonlinearity enters. It does not imply a spatially uniform deformation gradient: $\boldsymbol{F}(\boldsymbol{X})$ can vary throughout the body. This is distinct from calling the deformation map itself affine in $\boldsymbol{X}$, which would yield a constant $\boldsymbol{F}$.

A.4 Why may existence theory require different function spaces?

Answer. Square-integrable first derivatives do not automatically ensure that every finite-elasticity energy or weak form is finite. Membership in $H^1 = W^{1,2}$ means

$$\int_{\mathcal{B}_0} \left(|\boldsymbol{\varphi}|^2 + |\operatorname{Grad}\boldsymbol{\varphi}|^2\right) \mathrm{d}V < \infty.$$

This is natural for the quadratic energy of ordinary linear elasticity. In finite elasticity, the required integrability depends on the growth of the chosen energy.

For example, the Saint Venant–Kirchhoff energy is

$$\Psi(\boldsymbol{E}) = \frac{\lambda}{2}(\operatorname{tr}\boldsymbol{E})^2 + \mu\boldsymbol{E}:\boldsymbol{E}, \qquad \boldsymbol{E} = \frac{1}{2}(\boldsymbol{F}^T\boldsymbol{F} - \boldsymbol{I}).$$

With positive-definite elastic coefficients, this energy exhibits quartic growth as $|\boldsymbol{F}| \to \infty$. The condition $\boldsymbol{F} \in L^2$ does not imply $\boldsymbol{F} \in L^4$; $W^{1,4}$ is therefore natural for its energy integrability. This is an integrability observation, not an existence theorem for that material model.

The stress–test-gradient product must also be integrable. Let $p' = p/(p-1)$, so $1/p + 1/p' = 1$. On a bounded reference body, a stress-growth bound

$$|\boldsymbol{P}(\boldsymbol{F}, \boldsymbol{X})| \le c\big(1 + |\boldsymbol{F}|^{p-1}\big), \qquad \boldsymbol{F} \in L^p(\mathcal{B}_0), \quad 1 < p < \infty,$$

with c independent of $\boldsymbol{X}$, implies $\boldsymbol{P} \in L^{p'}(\mathcal{B}_0)$ because $(p-1)p' = p$. This is an assumption on the energy derivative; differentiability and p-growth of the energy alone do not imply it.

For $\operatorname{Grad}\boldsymbol{\eta} \in L^p(\mathcal{B}_0)$, pointwise tensor Cauchy–Schwarz followed by Hölder's inequality yields

$$\left|\int_{\mathcal{B}_0} \boldsymbol{P} : \operatorname{Grad}\boldsymbol{\eta}\, \mathrm{d}V\right| \le \int_{\mathcal{B}_0} |\boldsymbol{P}|\,|\operatorname{Grad}\boldsymbol{\eta}|\, \mathrm{d}V \le \|\boldsymbol{P}\|_{L^{p'}}\, \|\operatorname{Grad}\boldsymbol{\eta}\|_{L^p} < \infty.$$

Here $|\cdot|$ denotes the Frobenius norm, and all function norms are over $\mathcal{B}_0$. Thus Hölder pairs conjugate integrability exponents to bound the internal virtual work; it does not itself establish the stress-growth bound. See Hunter, Chapter 7, Theorem 7.7. For $p = 2$, this is the usual L^2 Cauchy–Schwarz estimate.

Additional admissibility conditions include $J > 0$ and, where required, global injectivity. Many energies also become singular as $J \to 0^+$; their stresses require additional control beyond the simple polynomial bound above. See Ball (2010), especially the discussion of stress integrability in Section 1.4.

Existence proofs require suitable hypotheses on the energy, loads, and admissible limits, such as coercivity and weak lower semicontinuity; see Ball (2010, Section 1.3). Simply choosing a larger Sobolev exponent is insufficient. The H^1 spaces in the main text provide an introductory conforming finite-element framework; the rigorous setting is material-dependent.

A.5 What does the composition defining spatial virtual displacement mean?

Question. How should $\boldsymbol{w} = \boldsymbol{\eta} \circ \boldsymbol{\varphi}^{-1}$ be interpreted?

Answer. The symbol $\circ$ denotes composition: apply the function on the right first. The reference and spatial descriptions use different position arguments,

$$\boldsymbol{\eta} : \mathcal{B}_0 \to \mathbb{R}^3, \qquad \boldsymbol{w} : \mathcal{B}_t \to \mathbb{R}^3.$$

At a current position $\boldsymbol{x}$, first find the material point $\boldsymbol{X} = \boldsymbol{\varphi}^{-1}(\boldsymbol{x})$, then evaluate its virtual displacement:

$$\boxed{\boldsymbol{w}(\boldsymbol{x}) = \boldsymbol{\eta}\big(\boldsymbol{\varphi}^{-1}(\boldsymbol{x})\big).}$$

Equivalently,

$$\boldsymbol{w}(\boldsymbol{\varphi}(\boldsymbol{X})) = \boldsymbol{\eta}(\boldsymbol{X}).$$

These expressions represent the same virtual displacement vector, labeled by current or reference position. In a fixed Cartesian basis, no additional multiplication by $\boldsymbol{F}$ is needed for the vector value: $\boldsymbol{\eta}$ already represents a variation of the current position. Composition changes its argument.

For example, in one dimension, let $x = \varphi(X) = 2X$ and $\eta(X) = \alpha X$. Then

$$w(x) = \eta(x/2) = \frac{\alpha x}{2}, \qquad w(2X) = \eta(X).$$

Differentiating $\boldsymbol{\eta}(\boldsymbol{X}) = \boldsymbol{w}(\boldsymbol{\varphi}(\boldsymbol{X}))$ using the chain rule yields

$$\mathrm{Grad}_X\, \boldsymbol{\eta} = (\nabla_x \boldsymbol{w})\boldsymbol{F}, \qquad \nabla_x \boldsymbol{w} = (\mathrm{Grad}_X\, \boldsymbol{\eta})\boldsymbol{F}^{-1}.$$

These identities relate the reference and spatial virtual-work integrals. A global inverse requires a one-to-one deformation; $J > 0$ alone only ensures local invertibility for a sufficiently smooth map.

A.6 Why is there a factor of two in the second Piola–Kirchhoff stress?

Question. Why does $\boldsymbol{S} = \partial\Psi/\partial\boldsymbol{E} = 2\,\partial\Psi/\partial\boldsymbol{C}$?

Answer. The factor arises directly from the definition of the Green–Lagrange strain:

$$\boldsymbol{E} = \frac{1}{2}(\boldsymbol{C} - \boldsymbol{I}), \qquad \delta\boldsymbol{E} = \frac{1}{2}\delta\boldsymbol{C}.$$

The stress $\boldsymbol{S}$ is work-conjugate to $\boldsymbol{E}$, so

$$\delta\Psi = \boldsymbol{S} : \delta\boldsymbol{E} = \frac{1}{2}\boldsymbol{S} : \delta\boldsymbol{C}.$$

Comparing this with $\delta\Psi = (\partial\Psi/\partial\boldsymbol{C}) : \delta\boldsymbol{C}$ for arbitrary symmetric variations yields

$$\boxed{\boldsymbol{S} = \frac{\partial\Psi}{\partial\boldsymbol{E}} = 2\frac{\partial\Psi}{\partial\boldsymbol{C}}.}$$

To distinguish the two representations explicitly, write

$$U(\boldsymbol{E}) = W(2\boldsymbol{E} + \boldsymbol{I}), \qquad \frac{\partial U}{\partial\boldsymbol{E}} = 2\frac{\partial W}{\partial\boldsymbol{C}}.$$

One must transform the argument of the energy, rather than simply rename it. A scalar example makes the coefficient explicit:

$$U(E) = \frac{1}{2}kE^2, \qquad E = \frac{C-1}{2}, \qquad W(C) = \frac{k}{8}(C-1)^2.$$

Thus

$$\frac{dU}{dE} = kE, \qquad 2\frac{dW}{dC} = \frac{k}{2}(C-1) = kE.$$

The factor 2 follows from the definition of $\boldsymbol{E}$. These formulas describe the unconstrained hyperelastic stress when energy is measured per reference volume. Exact incompressibility additionally introduces a pressure-constraint contribution to the total stress.

B Exercises for Hand Calculation

These six exercises link the formulas to physical interpretation and finite-element implementation. Work symbolically before substituting numbers; a calculator is optional. For each stress, traction, and integral, state the configuration and the measure used. Brief hints and answer checks are provided separately in Appendix C.

B.1 Rigid rotation, stretch, and objectivity

Focus: Sections 1–2 and Appendix A.1.

In two dimensions, consider $\boldsymbol{\varphi}(\boldsymbol{X}) = \boldsymbol{Q}_\theta \boldsymbol{U}\boldsymbol{X}$, where

$$\boldsymbol{Q}_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad \boldsymbol{U} = \begin{pmatrix} \lambda & 0 \\ 0 & 1 \end{pmatrix}, \qquad \lambda > 0.$$

Use $\boldsymbol{F} = \operatorname{Grad}\boldsymbol{\varphi}$, $\boldsymbol{E} = (\boldsymbol{F}^T\boldsymbol{F} - \boldsymbol{I})/2$, and $\boldsymbol{\varepsilon} = \operatorname{sym}\operatorname{Grad}\boldsymbol{u}$.

(a) Compute $\boldsymbol{F}$, J, $\boldsymbol{C}$, and $\boldsymbol{E}$ for arbitrary θ and λ. Which quantities change when only θ varies?

(b) Set $\lambda = 1$ and $\theta = \pi/2$. Compute $\operatorname{Grad}\boldsymbol{u}$ and $\boldsymbol{\varepsilon}$. Explain why the infinitesimal strain yields a misleading result for this rigid motion.

(c) Compare $\Psi_1 = \mu\boldsymbol{E} : \boldsymbol{E}$ and $\Psi_2 = \mu|\boldsymbol{F} - \boldsymbol{I}|^2$, where $\mu > 0$ and $|\cdot|$ denotes the Frobenius norm. Evaluate both at $\lambda = 3/2$ for $\theta = 0$ and $\theta = \pi/2$. Which energy is objective?

(d) Does objectivity prohibit preferred material directions? Distinguish this question from whether $\boldsymbol{F}$ is spatially constant in this example.

B.2 One stress state, two configurations, identical work

Focus: Sections 2, 7, and 10; Appendix A.5.

At a material point in a two-dimensional body, let

$$\boldsymbol{F} = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}, \qquad \boldsymbol{S} = s_0 \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \boldsymbol{G} = \operatorname{Grad}\boldsymbol{\eta} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix},$$

where $s_0 > 0$ is a stress scale. Interpret volume and surface measures per unit out-of-plane thickness. No constitutive law needs to be inferred from this prescribed stress state.

(a) Calculate J, $\boldsymbol{P} = \boldsymbol{F}\boldsymbol{S}$, and $\boldsymbol{\sigma} = J^{-1}\boldsymbol{P}\boldsymbol{F}^T$. Explain why $\boldsymbol{P}$ need not be symmetric even though $\boldsymbol{S}$ and $\boldsymbol{\sigma}$ are symmetric.

(b) On a reference face with $\boldsymbol{N} = (1,0)^T$, use Nanson's formula to find $\boldsymbol{n}$ and $\mathrm{d}a/\,\mathrm{d}A$. Verify $\boldsymbol{P}\boldsymbol{N}\,\mathrm{d}A = \boldsymbol{\sigma}\boldsymbol{n}\,\mathrm{d}a$.

(c) Calculate $\delta\boldsymbol{E} = \operatorname{sym}(\boldsymbol{F}^T\boldsymbol{G})$ and $\nabla_x\boldsymbol{w} = \boldsymbol{G}\boldsymbol{F}^{-1}$. Verify

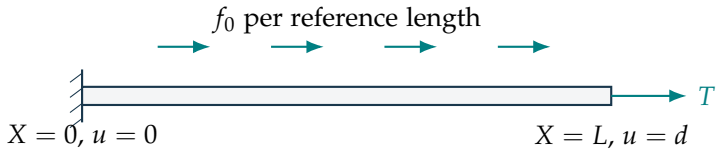
$$\boldsymbol{P} : \boldsymbol{G} = \boldsymbol{S} : \delta\boldsymbol{E} = J\boldsymbol{\sigma} : \nabla_x\boldsymbol{w}.$$

(d) Why would comparing $\boldsymbol{P} : \boldsymbol{G}$ directly with $\boldsymbol{\sigma} : \nabla_x\boldsymbol{w}$, without J, yield different values? Identify the physical quantities that should be compared.

B.3 A loaded bar: from strong form to one finite element

Focus: Sections 3–5 and 9.

A one-dimensional bar occupies $0 < X < L$ and has constant reference area A_0. Its axial displacement is $u(X)$, its stretch is $\lambda = 1 + u'(X) > 0$, and its nominal stress is $P(\lambda)$. Prescribe $u(0) = 0$, a signed dead force T at $X = L$, and a uniform body force f_0 per unit reference length, positive to the right. This is a one-dimensional constitutive model, not a traction-free reduction of a three-dimensional material.



(a) Starting from $(A_0P)' + f_0 = 0$ and $A_0P(L) = T$, multiply by η with $\eta(0) = 0$ and integrate by parts. Derive the weak form, keeping the boundary signs explicit.

(b) Use $u^h = (X/L)d$ and $\eta^h = (X/L)\delta d$. Derive the scalar residual $R(d) = f_{\text{int}} - f_{\text{ext}}$ and the two consistent nodal body-force contributions. Why is the body-force contribution at the free node $f_0L/2$, rather than f_0L?

(c) Integrate the strong equilibrium equation to obtain the exact nominal-stress distribution. For $f_0 \neq 0$, can the constant stress of one homogeneous linear element satisfy it pointwise? What does the discrete weak equation enforce instead?

(d) Recover the support reaction using the constrained-node residual. Verify equilibrium of the entire bar, including the distributed load.

B.4 An exact Newton tangent and its two contributions

Focus: Sections 10–11 and Appendix A.6.

For the bar in Exercise B.3, set $f_0 = 0$ and use the one-dimensional hyperelastic law

$$\Psi(E) = \tfrac{1}{2}KE^2, \qquad E = \tfrac{1}{2}(\lambda^2 - 1), \qquad K > 0.$$

Here, K is a stress modulus, distinct from the scalar element tangent K_T.

(a) Derive $S = \partial\Psi/\partial E$, $P = \partial\Psi/\partial\lambda$, $R(d)$, and $K_T = dR/dd$. Verify $P = \lambda S$ independently.

(b) Differentiate $P = \lambda S$ to identify the material contribution $K_{\text{mat}} = (A_0/L)K\lambda^2$ and the geometric contribution $K_{\text{geo}} = (A_0/L)S$. Do they sum to K_T?

(c) Let $T/(A_0K) = 15/16$ and start at $\lambda^{(0)} = 1$. Perform one full Newton step using the dimensionless residual. Check that $\lambda_* = 3/2$ is an equilibrium stretch. Does one Newton step reach it exactly?

(d) For which positive stretches is K_T negative? Explain why a positive modulus K and a positive material contribution do not guarantee a positive total tangent. Treat this as a feature of the specified toy model.

B.5 When does slow loading justify quasi-static equilibrium?

Focus: Section 8 and Appendix A.2.

Consider the small-amplitude, unprestressed limit of the same bar, with density ρ_0 and modulus K. Its displacement satisfies

$$\rho_0 u_{tt} = K u_{XX}.$$

Let $X = L\widehat{X}$, $t = T_{\text{load}}\widehat{t}$, and $u = U\widehat{u}$. Define the wave speed $c = \sqrt{K/\rho_0}$ and the transit time $T_{\text{wave}} = L/c$.

(a) Nondimensionalize the equation. Show that the coefficient multiplying $\widehat{u}_{\widehat{t}\widehat{t}}$ is $\epsilon = (T_{\text{wave}}/T_{\text{load}})^2$.

(b) Evaluate ϵ for smooth load ramps with durations $100T_{\text{wave}}$ and $T_{\text{wave}}/10$. Which is a plausible quasi-static regime, assuming initially negligible vibrations?

(c) Distinguish a held static state, a quasi-static loading path, and a Newton iteration at a fixed load. Is the Newton index a physical time variable?

(d) A very slow loading path reaches a limit point or a snap-through instability. Explain why the preceding timescale estimate need not justify quasi-static motion through the ensuing jump. Relate this to loss of tangent stiffness.

B.6 Finite energy and the stress–test-gradient pairing

Focus: Section 4 and Appendix A.4.

On the nondimensional interval $0 < X < 1$, consider the scalar model

$$\varphi(X) = \frac{X^{1-a}}{1-a}, \qquad F = \varphi'(X) = X^{-a}, \qquad \Psi(F) = \frac{F^4}{4}, \qquad P = \frac{d\Psi}{dF} = F^3, \qquad 0 < a < 1.$$

This example tests integrability; it is not claimed to solve a loaded equilibrium problem. Use $\int_0^1 X^{-q}\,\mathrm{d}X < \infty$ if and only if $q < 1$.

(a) Find the conditions on a for $\varphi \in H^1(0,1)$ and for finite stored energy. Use $a = 1/3$ to show that membership in H^1 does not suffice.

(b) Let $\eta(X) = X^{1-b}/(1-b)$ with $0 < b < 1/4$, so $\eta(0) = 0$ and $\eta' = X^{-b} \in L^4$. For $0 < a < 1/4$, verify $P \in L^{4/3}$ and compute $\int_0^1 P\eta'\,\mathrm{d}X$ exactly.

(c) Write the Hölder estimate for this integral and evaluate both norms. Why do the conjugate exponents $4/3$ and 4 appear?

(d) With $a = b = 1/8$, evaluate the integral and the product of norms. Are they equal? Finally, does finite energy plus $F > 0$ by itself establish the existence of an equilibrium minimizer?

C Hints and Answer Checks for the Exercises

Use these checks after attempting the exercises in Appendix B. They provide intermediate and final results; you should still write out the integration-by-parts steps, tensor products, and physical explanations.

Exercise B.1: rotation and objectivity

Hint. Form $\boldsymbol{F}^T\boldsymbol{F}$ before substituting the rotation angle.

$$\boldsymbol{F} = \boldsymbol{Q}_\theta \boldsymbol{U}, \qquad J = \lambda, \qquad \boldsymbol{C} = \operatorname{diag}(\lambda^2, 1), \qquad \boldsymbol{E} = \operatorname{diag}((\lambda^2 - 1)/2, 0).$$

Only $\boldsymbol{F}$ among these four quantities depends on θ. For a quarter-turn with $\lambda = 1$,

$$\operatorname{Grad} \boldsymbol{u} = \begin{pmatrix} -1 & -1 \\ 1 & -1 \end{pmatrix}, \qquad \boldsymbol{\varepsilon} = -\boldsymbol{I}, \qquad \boldsymbol{E} = \boldsymbol{0}.$$

The small-gradient assumption fails, even though the exact strain is zero. At $\lambda = 3/2$, $\Psi_1 = 25\mu/64$ for both angles. In contrast, $\Psi_2 = \mu/4$ at $\theta = 0$ and $21\mu/4$ at $\theta = \pi/2$. Thus, Ψ_1 is objective and Ψ_2 is not. Objectivity permits anisotropy. The constant $\boldsymbol{F}$ here follows from the chosen affine motion, not from objectivity.

Exercise B.2: stress, traction, and virtual work

Hint. Nanson's formula determines both the current unit normal and the surface-measure ratio.

$$J = 2, \qquad \boldsymbol{P} = s_0 \begin{pmatrix} 7 & 4 \\ 1 & 2 \end{pmatrix}, \qquad \boldsymbol{\sigma} = s_0 \begin{pmatrix} 9 & 2 \\ 2 & 1 \end{pmatrix}.$$

The first Piola stress relates spatial force to reference area; symmetry is not generally required for this two-configuration tensor. On the specified face,

$$\boldsymbol{n} = \frac{(1,-1)^T}{\sqrt{2}}, \qquad \frac{\mathrm{d}a}{\mathrm{d}A} = \sqrt{2}, \qquad \boldsymbol{P}\boldsymbol{N} = s_0(7,1)^T, \qquad \boldsymbol{\sigma}\boldsymbol{n} = \frac{s_0}{\sqrt{2}}(7,1)^T.$$

The work quantities are

$$\delta\boldsymbol{E} = \begin{pmatrix} 2 & 1/2 \\ 1/2 & 0 \end{pmatrix}, \qquad \nabla_x \boldsymbol{w} = \begin{pmatrix} 1/2 & -1/2 \\ 0 & 0 \end{pmatrix},$$

$$\boldsymbol{P} : \boldsymbol{G} = \boldsymbol{S} : \delta\boldsymbol{E} = 7s_0, \qquad \boldsymbol{\sigma} : \nabla_x \boldsymbol{w} = \frac{7}{2} s_0.$$

These densities are per different volumes. The physical virtual work agrees because $\mathrm{d}v = 2\,\mathrm{d}V$; likewise, the traction forces agree on corresponding surface patches.

Exercise B.3: bar equilibrium and discretization

Hint. Retain $[A_0 P\eta]_0^L$ until the essential and natural boundary data have been applied.

$$\int_0^L A_0 P\eta'\,\mathrm{d}X = T\eta(L) + \int_0^L f_0\eta\,\mathrm{d}X, \qquad R(d) = A_0 P(1+d/L) - T - \frac{f_0 L}{2}.$$

The linear shape functions $N_1 = 1 - X/L$ and $N_2 = X/L$ yield consistent body-force contributions of $f_0 L/2$ at each node. The free equation includes only the second; the first contributes to the reaction at the constrained node.

$$P_{\text{exact}}(X) = \frac{T + f_0(L-X)}{A_0}, \qquad P^h = \frac{T + f_0 L/2}{A_0} \quad \text{at discrete equilibrium.}$$

For $f_0 \neq 0$, the constant P^h cannot satisfy $(A_0 P^h)' + f_0 = 0$ pointwise. It equals the reference-length average of the exact stress and balances the residual against the available finite-element test field. This does not make the element displacement exact. With the convention that the reaction is the force exerted by the support on the bar,

$$R_{\text{support}} = -A_0 P^h - \frac{f_0 L}{2} = -T - f_0 L.$$

Thus $R_{\text{support}} + T + f_0 L = 0$, as required.

Exercise B.4: consistent tangent and Newton step

Hint. Differentiate with respect to d using $d\lambda/dd = 1/L$.

$$S = \frac{K}{2}(\lambda^2 - 1), \quad P = \lambda S = \frac{K}{2}\lambda(\lambda^2 - 1), \quad R = A_0 P - T, \quad K_T = \frac{A_0 K}{2L}(3\lambda^2 - 1).$$

The product rule yields

$$K_T = \underbrace{\frac{A_0 K}{L}\lambda^2}_{K_{\text{mat}}} + \underbrace{\frac{A_0}{L} S}_{K_{\text{geo}}}.$$

For $r(\lambda) = \frac{1}{2}\lambda(\lambda^2 - 1) - 15/16$, one has $r(1) = -15/16$ and $r'(1) = 1$. Hence

$$\lambda^{(1)} = 1 - \frac{r(1)}{r'(1)} = \frac{31}{16}, \qquad \lambda_* = \frac{3}{2}.$$

The first step overshoots; a nonlinear residual is not generally eliminated in one iteration. The total tangent is negative for $0 < \lambda < 1/\sqrt{3}$, despite $K_{\text{mat}} > 0$. The compressive initial stress contributes a negative geometric term. Positive strain-energy curvature with respect to E does not guarantee positive curvature with respect to the nodal displacement.

Exercise B.5: quasi-static scaling

Hint. Compare the characteristic acceleration U/T_{load}^2 with the elastic term KU/L^2.

$$\epsilon \widehat{u}_{\widehat{t}\widehat{t}} = \widehat{u}_{\widehat{X}\widehat{X}}, \qquad \epsilon = \frac{\rho_0 L^2}{K T_{\text{load}}^2} = \left(\frac{T_{\text{wave}}}{T_{\text{load}}}\right)^2.$$

The two values are 10^{-4} and 100, respectively. With smooth loading and negligible initial vibration, the first is a plausible quasi-static regime; the second is not. A static state is time-independent; a quasi-static path evolves with inertia neglected; and Newton iterations solve a fixed-load algebraic equilibrium problem. The iteration index is not physical time.

Near an instability, a relevant tangent stiffness can approach zero, and the associated structural response time grows. A rapid jump may follow even under slow loading. The estimate based on the initial unprestressed modulus does not capture that jump; dynamics or a suitable equilibrium-path continuation analysis may be needed, depending on the question being asked.

Exercise B.6: integrability and Hölder

Hint. Test the powers F^2, F^4, $|P|^{4/3}$, and $|\eta'|^4$ separately before multiplying the fields.

Since φ is bounded for $0 < a < 1$, membership in H^1 is equivalent to $2a < 1$. Finite stored energy requires $4a < 1$. Thus, $a = 1/3$ yields an H^1 deformation with infinite quartic energy. For $a, b < 1/4$,

$$P = X^{-3a} \in L^{4/3}, \qquad \eta' = X^{-b} \in L^4, \qquad \int_0^1 P\eta' \, \mathrm{d}X = \frac{1}{1-3a-b}.$$

Because $3a + b < 1$, the product is integrable. Hölder's inequality yields the quantitative estimate

$$\frac{1}{1-3a-b} \leq \|P\|_{L^{4/3}} \|\eta'\|_{L^4} = (1-4a)^{-3/4}(1-4b)^{-1/4}.$$

The exponents are conjugate because $3/4 + 1/4 = 1$. At $a = b = 1/8$, both sides equal 2: the equality condition is met because $|P|^{4/3}$ and $|\eta'|^4$ coincide. Finite energy and $F > 0$ alone do not prove the existence of an equilibrium minimizer; the energy, loading, boundary data, and admissible limiting deformations must satisfy additional necessary hypotheses.

D Multiple-Choice Concept Quiz

Choose **one best answer** to each question. Work without the answer key first, and explain to yourself why each alternative fails. The notation follows the notes; Ψ denotes stored energy per reference volume, and vectors use fixed Cartesian bases. Answers and brief explanations appear in Appendix E.

1. A body undergoes a uniform translation $\boldsymbol{u}(\boldsymbol{X}) = \boldsymbol{c}$, where $\boldsymbol{c}$ is a constant. What is its deformation gradient?

A. $\boldsymbol{F} = \boldsymbol{0}$.
B. $\boldsymbol{F} = \boldsymbol{I}$.
C. $\boldsymbol{F} = \boldsymbol{I} + \boldsymbol{c} \otimes \boldsymbol{c}$.
D. $\boldsymbol{F}$ cannot be determined without the magnitude of $\boldsymbol{c}$.

2. Why is $\boldsymbol{F} = \boldsymbol{I} + \operatorname{Grad} \boldsymbol{u}$ affine in $\operatorname{Grad} \boldsymbol{u}$?

A. It requires $\boldsymbol{F}$ to be spatially uniform.
B. It makes the stress linear in the displacement.
C. It neglects all quadratic terms in the Green–Lagrange strain.
D. It is a linear dependence plus a fixed offset, yielding $\Delta \boldsymbol{F} = \operatorname{Grad} \Delta \boldsymbol{u}$.

3. For a three-dimensional deformation, $\boldsymbol{F} = \operatorname{diag}(2, 1/2, 1)$. Which statement is correct?

A. Local volume is unchanged, even though the shape changes.
B. Every material line retains its length.
C. The Green–Lagrange strain vanishes because $J = 1$.
D. The motion is a rigid rotation because $\det \boldsymbol{F} = 1$.

4. A body undergoes a finite rigid rotation $\boldsymbol{\varphi} = \boldsymbol{Q}\boldsymbol{X}$, with $\boldsymbol{Q}^T\boldsymbol{Q} = \boldsymbol{I}$ and $\det \boldsymbol{Q} = 1$. Which statement about strain is correct?

A. Both $\boldsymbol{E}$ and $\operatorname{sym}(\operatorname{Grad} \boldsymbol{u})$ must vanish for every rotation.
B. $\boldsymbol{E}$ is nonzero whenever $\boldsymbol{Q} \neq \boldsymbol{I}$.
C. $\boldsymbol{E} = \boldsymbol{0}$, but $\operatorname{sym}(\operatorname{Grad} \boldsymbol{u})$ need not vanish.
D. $\boldsymbol{E} = \boldsymbol{Q} - \boldsymbol{I}$.

5. What does objectivity require of a hyperelastic stored-energy law under a superposed rigid rotation $\boldsymbol{Q}$?

A. $\Psi(\boldsymbol{Q}\boldsymbol{F}) = \Psi(\boldsymbol{F})$, and the spatial stress components remain unchanged.
B. $\Psi(\boldsymbol{Q}\boldsymbol{F}) = \Psi(\boldsymbol{F})$, while $\boldsymbol{\sigma}$ transforms as $\boldsymbol{Q}\boldsymbol{\sigma}\boldsymbol{Q}^T$.
C. $\Psi(\boldsymbol{Q}\boldsymbol{F}) = \Psi(\boldsymbol{F})$ only if the material is isotropic.
D. Ψ may depend only on $J = \det \boldsymbol{F}$.

Concept quiz: questions 6–10

6. Can a fiber-reinforced hyperelastic material be objective and anisotropic?

A. Yes; observer independence and dependence on material directions are compatible.
B. No; objectivity requires the response to be identical in every material direction.
C. Only if the fibers carry no stress.
D. Only in the infinitesimal-displacement limit.

7. Which pairing correctly identifies traction and the area it acts on?

A. $\boldsymbol{PN}$: force per current area; $\boldsymbol{\sigma n}$: force per reference area.
B. Both $\boldsymbol{PN}$ and $\boldsymbol{\sigma n}$: force per current area.
C. Both $\boldsymbol{PN}$ and $\boldsymbol{\sigma n}$: force per reference area.
D. $\boldsymbol{PN}$: force per reference area; $\boldsymbol{\sigma n}$: force per current area.

8. In a classical continuum without couple stresses, which statement regarding stress symmetry is generally correct?

A. $\boldsymbol{P}$, $\boldsymbol{S}$, and $\boldsymbol{\sigma}$ are all symmetric.
B. Only $\boldsymbol{P}$ is necessarily symmetric.
C. $\boldsymbol{S}$ and $\boldsymbol{\sigma}$ are symmetric; $\boldsymbol{P}$ need not be.
D. None of the stress tensors can be symmetric at finite deformation.

9. What is the physical meaning of $\boldsymbol{PN}\ \mathrm{d}A = \boldsymbol{\sigma n}\ \mathrm{d}a$ for corresponding surface patches?

A. The two descriptions yield the same physical force vector.
B. The nominal and Cauchy traction vectors are always numerically equal.
C. The reference and current unit normals must coincide.
D. The reference and current surface areas must be equal.

10. Why does $\boldsymbol{S} = \partial\Psi/\partial\boldsymbol{E} = 2\,\partial\Psi/\partial\boldsymbol{C}$ include the factor 2?

A. Each shear component has two symmetric off-diagonal entries.
B. Two configurations contribute equal amounts of stored energy.
C. The factor converts reference-volume energy to current-volume energy.
D. $\boldsymbol{E} = (\boldsymbol{C} - \boldsymbol{I})/2$, so $\delta\boldsymbol{C} = 2\delta\boldsymbol{E}$.

Concept quiz: questions 11–15

11. How does a quasi-static loading path differ from a static state?

A. A quasi-static path requires the displacement to remain constant.
B. A quasi-static path retains inertia but neglects elastic forces.
C. A quasi-static path may evolve with time while neglecting inertia; a static state is time-independent.
D. A quasi-static path is defined by performing many Newton iterations.

12. Which is the reference-configuration balance of linear momentum when $\boldsymbol{b}$ is the body force per unit mass?

A. $\operatorname{Div}\boldsymbol{P} + \rho_0\boldsymbol{b} = \rho\,\ddot{\boldsymbol{\varphi}}$.
B. $\operatorname{Div}\boldsymbol{P} + \rho_0\boldsymbol{b} = \rho_0\ddot{\boldsymbol{\varphi}}$.
C. $\operatorname{Div}\boldsymbol{P} + \rho_0\boldsymbol{b} = \rho_0\dot{\boldsymbol{\varphi}}$.
D. $\operatorname{Div}\boldsymbol{P} + \rho_0\boldsymbol{b} = \boldsymbol{0}$ even when inertia is retained.

13. For conservative loading, an equilibrium renders the total potential energy stationary. What follows from stationarity alone?

A. The equilibrium is a strict local minimum.
B. The equilibrium is unique.
C. The tangent stiffness is positive definite.
D. The first variation vanishes; stability requires further examination.

14. A prescribed nominal traction is defined on a fixed reference boundary and is independent of the nodal displacement $\boldsymbol{d}$. Its contribution to the external load tangent is:

A. Zero, even though its external force vector can be nonzero.
B. Equal to its external force vector.
C. Nonzero whenever the body deforms.
D. Equal to the geometric stiffness.

15. If $w = \eta \circ \boldsymbol{\varphi}^{-1}$ is the spatial virtual displacement, what does this mean at $\boldsymbol{x} = \boldsymbol{\varphi}(\boldsymbol{X})$?

A. $w(\boldsymbol{x}) = \boldsymbol{F}(\boldsymbol{X})\eta(\boldsymbol{X})$.
B. $w(\boldsymbol{x}) = \eta(\boldsymbol{X})$: the same vector is labeled by current position.
C. $w(\boldsymbol{x}) = \boldsymbol{\varphi}^{-1}(\eta(\boldsymbol{X}))$.
D. $w(\boldsymbol{x}) = \eta(\boldsymbol{X}) - \boldsymbol{u}(\boldsymbol{X})$.

Concept quiz: questions 16–20

16. Given the same shape functions in $\boldsymbol{X} = \sum_A N_A \boldsymbol{X}_A$ and $\boldsymbol{u} = \sum_A N_A \boldsymbol{d}_A$, which expression represents the consistent discrete deformation?

A. $\boldsymbol{\varphi} = \sum_A N_A \boldsymbol{d}_A$.
B. $\boldsymbol{\varphi} = \sum_A N_A (\boldsymbol{X}_A - \boldsymbol{d}_A)$.
C. $\boldsymbol{\varphi} = \sum_A N_A (\boldsymbol{X}_A + \boldsymbol{d}_A)$.
D. $\boldsymbol{\varphi} = \boldsymbol{F} \sum_A N_A \boldsymbol{d}_A$ for every element.

17. Which statement correctly describes a consistent Newton tangent for the discrete equilibrium equations?

A. It is the derivative of the residual being solved; local quadratic convergence requires additional hypotheses.
B. It guarantees convergence from every initial guess.
C. It is any symmetric approximation of the stiffness.
D. It makes the nonlinear residual vanish after a single iteration.

18. An implementation uses the exact $\mathbb{A} = \partial \boldsymbol{P} / \partial \boldsymbol{F}$ in its total Lagrangian internal tangent. Should it include a separate geometric-stiffness term?

A. Yes, because differentiating $\boldsymbol{P}$ captures only the material response.
B. Yes, but only if the material is hyperelastic.
C. No; the exact $\boldsymbol{P}$–$\boldsymbol{F}$ tangent already includes the geometric contribution.
D. No, because geometric stiffness vanishes for all elastic materials.

19. Let $\boldsymbol{R} = \boldsymbol{f}_{\text{int}} - \boldsymbol{f}_{\text{ext}}$. A pressure of constant magnitude acts normal to a deforming surface. Which tangent is consistent?

A. $\boldsymbol{K}_T = \partial \boldsymbol{f}_{\text{int}} / \partial \boldsymbol{d}$ because the pressure magnitude is constant.
B. $\boldsymbol{K}_T = \partial \boldsymbol{f}_{\text{int}} / \partial \boldsymbol{d} + \partial \boldsymbol{f}_{\text{ext}} / \partial \boldsymbol{d}$.
C. $\boldsymbol{K}_T = -\partial \boldsymbol{f}_{\text{ext}} / \partial \boldsymbol{d}$ only.
D. $\boldsymbol{K}_T = \partial \boldsymbol{f}_{\text{int}} / \partial \boldsymbol{d} - \partial \boldsymbol{f}_{\text{ext}} / \partial \boldsymbol{d}$.

20. What justifies the standard approximation $\boldsymbol{E} \approx \operatorname{sym}(\operatorname{Grad} \boldsymbol{u})$ in a fixed reference frame?

A. Small $\boldsymbol{E}$ alone, even with arbitrarily large rigid rotations.
B. Small $\| \operatorname{Grad} \boldsymbol{u} \|$, rendering the quadratic gradient term negligible.
C. $J = 1$, regardless of shear or rotation.
D. A linear law $\boldsymbol{S} = \mathbb{C} : \boldsymbol{E}$, regardless of the deformation.

E Quiz Answers and Brief Explanations

Each explanation identifies the principle that rules out the alternatives based on misconceptions.

1. B. A translation has $\operatorname{Grad} \boldsymbol{u} = \boldsymbol{0}$, so $\boldsymbol{F} = \boldsymbol{I}$. Displacement measures a change in position; its gradient captures relative motion between neighboring particles.

2. D. An affine map has a constant offset, here $\boldsymbol{I}$. Its increment is linear, but $\boldsymbol{F}$ may vary with position, and the stress and strain may still depend nonlinearly on the displacement.

3. A. $J = \det \boldsymbol{F} = 1$ preserves local volume. Yet $\boldsymbol{C} = \operatorname{diag}(4, 1/4, 1) \neq \boldsymbol{I}$, so lengths and shape change. Volume preservation does not imply rigidity.

4. C. $\boldsymbol{C} = \boldsymbol{Q}^T \boldsymbol{Q} = \boldsymbol{I}$, so $\boldsymbol{E} = \boldsymbol{0}$. The linearized strain is $\operatorname{sym}(\boldsymbol{Q} - \boldsymbol{I})$, which is generally nonzero for finite rotations and is unsuitable for large rotations.

5. B. Objectivity makes the scalar energy invariant under a superposed rigid motion. Spatial tensor components transform under the rotation; invariance of energy does not imply unchanged stress components in fixed axes.

6. A. Objectivity concerns superposed spatial rigid motions; isotropy concerns symmetry with respect to material direction. For example, a reference fiber direction $\boldsymbol{a}_0$ can enter via $\boldsymbol{a}_0 \cdot \boldsymbol{C}\boldsymbol{a}_0$, which is unchanged by a superimposed spatial rotation.

7. D. Nominal traction $\boldsymbol{P}\boldsymbol{N}$ uses the reference area; Cauchy traction $\boldsymbol{\sigma}\boldsymbol{n}$ uses the current area. They describe the same force on corresponding surface patches, but with different area measures.

8. C. Angular momentum balance implies that $\boldsymbol{\sigma}$ is symmetric, and $\boldsymbol{S} = J\boldsymbol{F}^{-1}\boldsymbol{\sigma}\boldsymbol{F}^{-T}$ is symmetric. The two-point tensor $\boldsymbol{P} = \boldsymbol{F}\boldsymbol{S}$ is generally nonsymmetric.

9. A. Nanson's relation transforms the oriented area, ensuring equal forces. The traction vectors need not be equal because the corresponding patch areas can differ and the normals can rotate.

10. D. The chain rule yields $\delta\Psi = (\partial\Psi/\partial\boldsymbol{C}) : 2\delta\boldsymbol{E} = \boldsymbol{S} : \delta\boldsymbol{E}$. The factor arises from the tensor definition of $\boldsymbol{E}$, not from an engineering-shear convention; Ψ is per reference volume.

Answers: questions 11–20

11. C. Quasi-static analysis follows equilibria as the load changes, with inertia neglected. Motion can occur along that path. A Newton iteration index tracks an algebraic solve and is not physical time.

12. B. All terms are forces per unit reference volume, so inertia uses $\rho_0 \ddot{\boldsymbol{\varphi}}$. The current density ρ appears in the spatial balance and satisfies $\rho J = \rho_0$; velocity is not acceleration.

13. D. A stationary point can be a minimum, a maximum, or a saddle. Stationarity implies equilibrium, whereas stability requires examining admissible perturbations and typically the second variation; uniqueness does not follow.

14. A. A fixed reference load vector has zero derivative with respect to $\boldsymbol{d}$. A nonzero force does not imply a nonzero load tangent; the issue is its dependence on the unknown deformation.

15. B. Composition means first finding the particle $\boldsymbol{X} = \boldsymbol{\varphi}^{-1}(\boldsymbol{x})$, then evaluating $\boldsymbol{\eta}$ at that point. This changes the argument, not the virtual-displacement vector; no multiplication by $\boldsymbol{F}$ is required.

16. C. The identity $\boldsymbol{\varphi} = \boldsymbol{X} + \boldsymbol{u}$ directly yields the stated interpolation. The current nodal position is $\boldsymbol{X}_A + \boldsymbol{d}_A$; the nodal displacement alone is not a position.

17. A. Consistency means $\boldsymbol{K}_T = \partial \boldsymbol{R} / \partial \boldsymbol{d}$. Local quadratic convergence also requires sufficient smoothness, a nonsingular solution tangent, a sufficiently close initial guess, and full Newton steps; it is not a global guarantee.

18. C. Applying the product rule to $\boldsymbol{P} = \boldsymbol{F}\boldsymbol{S}$ yields $\Delta \boldsymbol{P} = (\Delta \boldsymbol{F})\boldsymbol{S} + \boldsymbol{F}\Delta \boldsymbol{S}$. These yield the geometric and material contributions; adding geometry again would double-count it.

19. D. Differentiate both terms of the residual, keeping their signs. Even at a constant magnitude of pressure, the normal and surface area generally change, so the resulting external force can depend on deformation.

20. B. $\boldsymbol{E} = \operatorname{sym}(\operatorname{Grad} \boldsymbol{u}) + \frac{1}{2}(\operatorname{Grad} \boldsymbol{u})^T \operatorname{Grad} \boldsymbol{u}$. Small displacement gradients justify neglecting the quadratic term. A small finite strain alone can coexist with large rotations and does not justify this linearization.